GOVERNMENT ENTERPRISE SPECIFICATION

08 Enterprise Implementation Roadmap

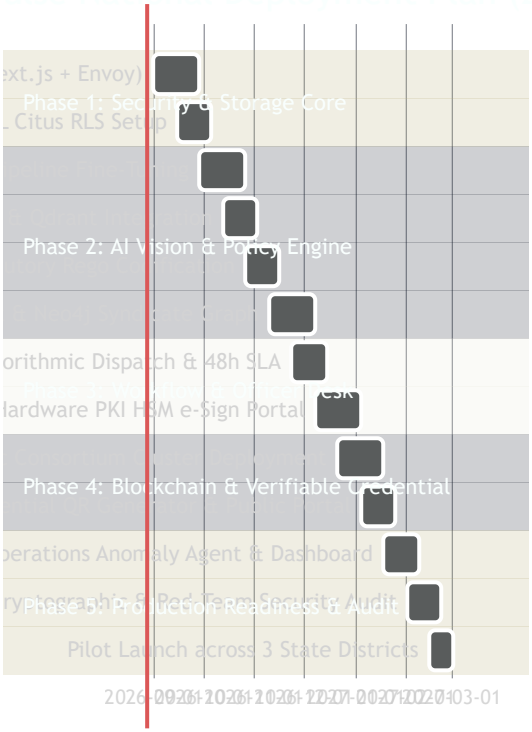
24-Week Rollout, K8s Sizing, HA/DR & Compliance • GovPulse Technical Architecture Suite • Version 1.0 (Production)

Enterprise Implementation Roadmap, Infrastructure Sizing & Compliance

Project: GovPulse Autonomous Digital Government Platform

---

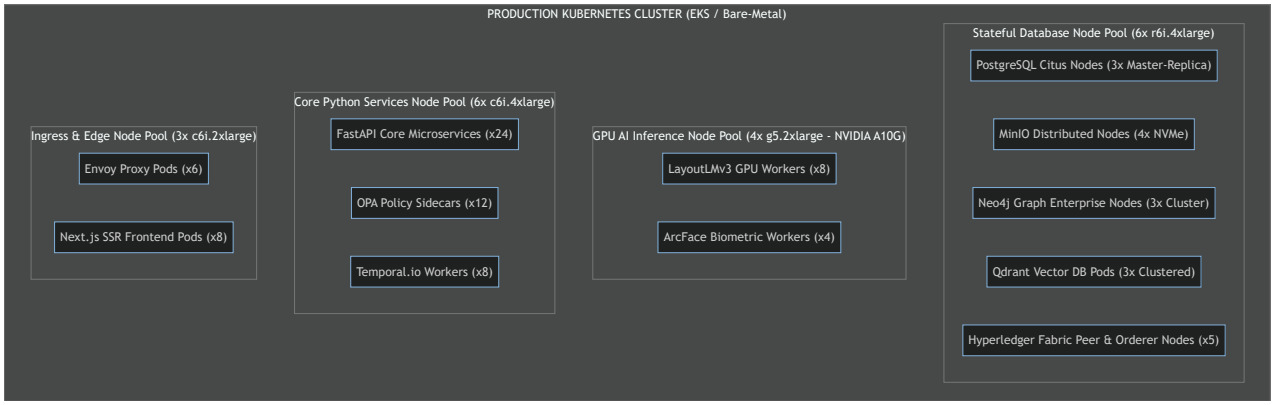
1. 24-Week Phased Implementation Roadmap



---

2. Production Infrastructure & Kubernetes (K8s) Cluster Sizing

To handle **10,000 concurrent submissions / min** with sub-minute AI turnaround and 99.99% availability:



---

3. High Availability, Disaster Recovery (DR) & RTO/RPO

METRIC	TARGET SLA	IMPLEMENTATION STRATEGY
High Availability (HA)	**99.99%** (< 52 min annual downtime)	Multi-Availability-Zone (Multi-AZ) active-active deployment across 3 independent data centers.
Recovery Point Objective (RPO)	**< 0 Seconds (Zero Data Loss)**	Synchronous replication on PostgreSQL WAL + Raft consensus across Hyperledger Fabric ordering peers.
Recovery Time Objective (RTO)	**< 60 Seconds**	Automated Kubernetes pod failover, AWS Route 53 health-check DNS failover, and Envoy circuit breakers.

---

4. Government Compliance & Regulatory Adherence

```
REGULATORY COMPLIANCE MATRIX
```

1. Zero PII On-Chain: Complete GDPR / Right-to-be-Forgotten Compliance
2. ISO/IEC 27001: Information Security Management System (ISMS) Standards
3. SOC 2 Type II: Certified Security, Confidentiality, and Processing Integrity
4. NIST SP 800-207: Complete Zero-Trust Architecture Alignment
5. W3C Verifiable Credentials Data Model v2.0: Global Interoperability
6. FIPS 140-3 Level 3: Hardware Security Module (HSM) Cryptographic Key Protection

---

5. Security Red-Team & Cryptographic Audit Strategy

Prior to public launch, the system undergoes three mandatory verification gates:

1. **Automated Dynamic & Static Security Scanning**:

- SAST (SonarQube / Bandit for Python), DAST (OWASP ZAP), Dependency scanning (Trivy / Snyk).

2. **Independent Cryptographic Audit**:

- Verification of SHA-256 state hash mathematical generation, Merkle tree construction, ECDSA signature non-malleability, and HSM PKCS#11 key isolation.

3. **Black-Box Penetration Testing**:

- Simulated adversarial attacks: Token theft, forged OCR document injection, collusion attempts to bypass blind dispatching, and DDoS stress testing up to 50,000 req/sec.